

Year 2 Assessed Problems

Semester 2

Assessed Problems 5

SOLUTIONS TO BE SUBMITTED

ON CANVAS BY

Wednesday 4th March 2026

at 17:00

Assessed problem sheet for Week 6

1. You are performing a scattering experiment, accelerating projectiles onto target nuclei. For the purpose of calculation, you may assume all nuclei to have infinite mass. You have two detectors measuring scattered particles covering the same amount of solid angle. One is placed at a scattering angle of 30° , one at 150° .
 - (a) You accelerate alpha particles with a kinetic energy of 30MeV at the target. Assuming the nucleus is point-like, what is the ratio of counts (i.e. scattered particles arriving) in the two detectors you would expect? [1]
 - (b) Assuming again a point-like nucleus. What is the ratio of counts you would expect when using 30MeV electrons instead? (Hint: If you can argue that $\beta \approx 1$, this makes the calculation simpler) [1]
 - (c) What is the value of $|q|$ in the case of scattering electrons at 150° ? [1]
 - (d) The nucleus turns out not to be point like. You measure the electron scattering at many different angles and produce a measured differential cross section $\frac{d\sigma}{d\Omega}(q^2)$. How can you connect this to the (electric) form factor of the nucleus? [1]
 - (e) The form factor turns out to be a Gaussian distribution. What does this tell you about the charge distribution in the nucleus? [1]

Y2 Neutrino Physics

Assessed problem sheet 2 (weeks 3–5)

1. The *radiation length* X_0 of a material can be defined as $7/9$ of the mean free path of a high energy photon before absorption via e^+e^- pair production. The radiation length for tungsten is $X_0 = 0.35$ cm. Compute the thickness of a tungsten sheet absorbing 90% of the incoming photons. [2]

2. Similarly to electron anti-neutrinos, muon anti-neutrinos can be detected using the inverse beta decay reaction: $\bar{\nu}_\mu p \rightarrow \mu^+ n$. Assuming that the proton is at rest, compute the threshold (i.e. the minimal $\bar{\nu}_\mu$ energy) for this reaction. [1]

3. The Cowan–Reines experiment discovered the electron anti-neutrino ($\bar{\nu}_e$) via the inverse beta-decay reaction $\bar{\nu}_e p \rightarrow e^+ n$, by detecting the positron via its annihilation with an atomic electron: $e^+e^- \rightarrow \gamma\gamma$. Is it possible to observe ν_e , ν_μ , and $\bar{\nu}_\mu$ via an inverse beta-decay reaction with a nucleon (p or n) followed by annihilation of the resulting lepton with an atomic e^- ? Justify your answer for each case. [2]

Assessed Problems 3

Richard Mason

Problem 1. Consider the two matrices

$$A = \begin{pmatrix} 1 & -2 & 0 \\ -2 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \quad B = \begin{pmatrix} 4 & -3 & 0 \\ -3 & 4 & 0 \\ 0 & 0 & 2 \end{pmatrix}$$

1. Find the (normalised) eigenvectors and eigenvalues for the matrix A . [4]
2. Show that the eigenvectors of A are also eigenvectors of B , and determine the eigenvalues. [2]
3. Calculate the commutator between A and B . [2]
4. Show that, in general, two commuting Hermitian matrices may be simultaneously diagonalised.
Hint: Write $A = UDU^{-1}$ and $B = UEU^{-1}$ where D and E are diagonal, and show that $AB = BA$ by considering whether diagonal matrices commute. [2]

Assessed Problems 2

Richard Mason
Foundations of Data Science

Problem 2. A probability density is given by

$$P(x|a) = \frac{x}{a^2} e^{-x/a} \quad x \geq 0$$

1. Calculate the MLE for the parameter a if N data points are i.i.d. drawn from this distribution. [5]
2. Calculate the Fisher information for the distribution,

$$I(a) = - \left\langle \frac{d^2}{da^2} \log P(x|a) \right\rangle$$

[2]

3. Employing this, calculate an *asymptotic* 0.95 confidence interval for the parameter a . You may simply use a_{MLE} as the estimate value. [3]

Note that for $x \sim \mathcal{N}(0, 1)$ then

$$\text{Prob}[-1.96 \leq x \leq 1.96] = 0.95$$

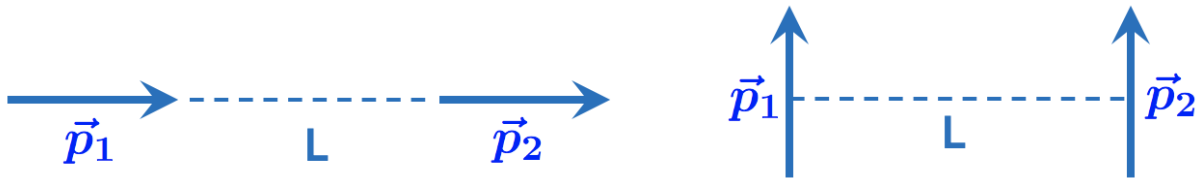
and the asymptotic distribution of the MLE is

$$a_{\text{MLE}} \sim \mathcal{N}\left(a, \frac{1}{NI(a)}\right)$$

Y2 Electromagnetism 2

Assessed problem sheet 2 (weeks 3–5)

1) Consider two parallel point-like electric dipoles with dipole moments $\vec{p}_1$ and $\vec{p}_2$, separated by a distance L (such that $L \gg a_{1,2}$). Consider the two cases shown below. In each case, find the magnitude and direction of the electrostatic force acting on each of the dipoles. [3]



2) Electric susceptibility χ_E of gaseous polar dielectric material at a fixed density corresponding to molar volume of 22.4 litres/mole is measured to be 0.004 at the temperature of 300 K, and 0.0028 at 500 K. Compute the permanent electric dipole moment of a molecule. [3]

3) Assume that the electron is a solid sphere of radius 2.8 fm with electric charge distributed uniformly over the volume, and that the electron dipole magnetic moment arises from rotation about an axis of symmetry. Find the angular velocity of rotation, ω , that gives rise to a magnetic moment equal to the Bohr magneton: $\mu_B = 9.3 \times 10^{-24} \text{ Am}^2$. [4]
